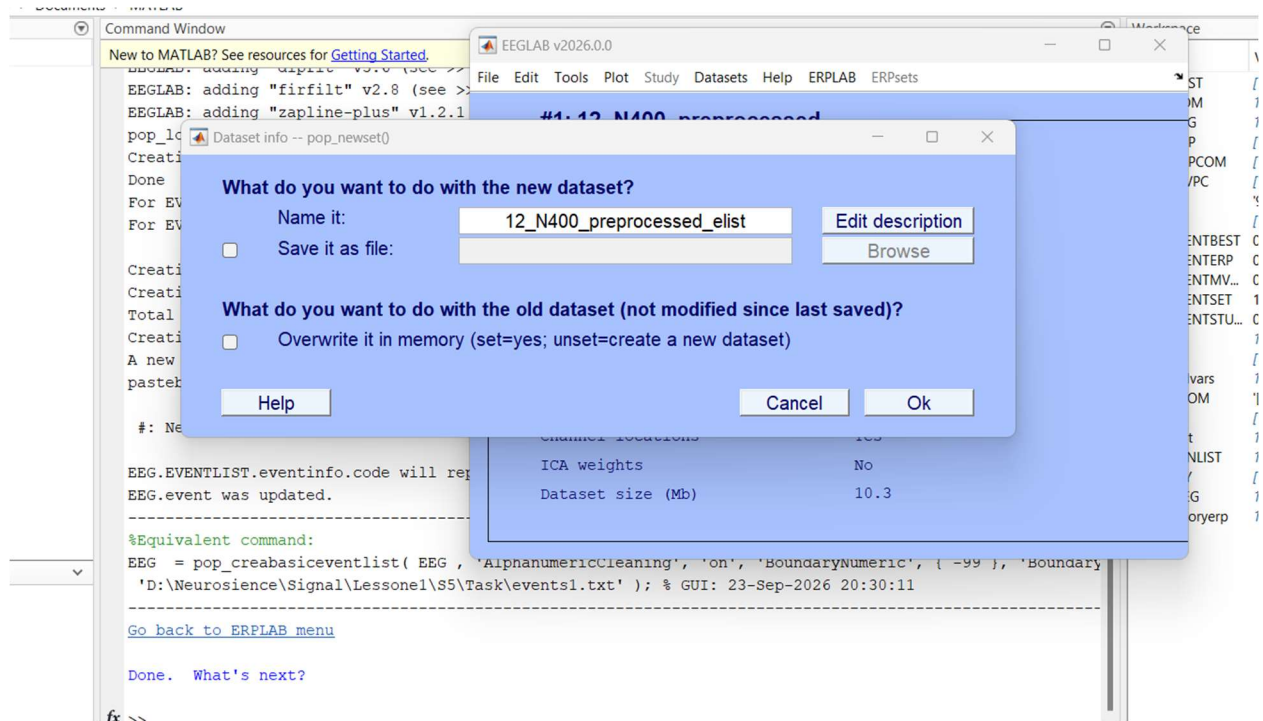


Assignment5

Name: Hamid Reza Ghahramani

You have one participant from the ERP CORE N400 Experiment (**12_N400_preprocessed.set**) along with the corresponding Bin Descriptor File (**BDF_N400.txt**). Please complete the following steps, take screenshots of each step, and compile them into a single PDF report.

1. Create the Eventlist of event codes and save it as *events.txt*.



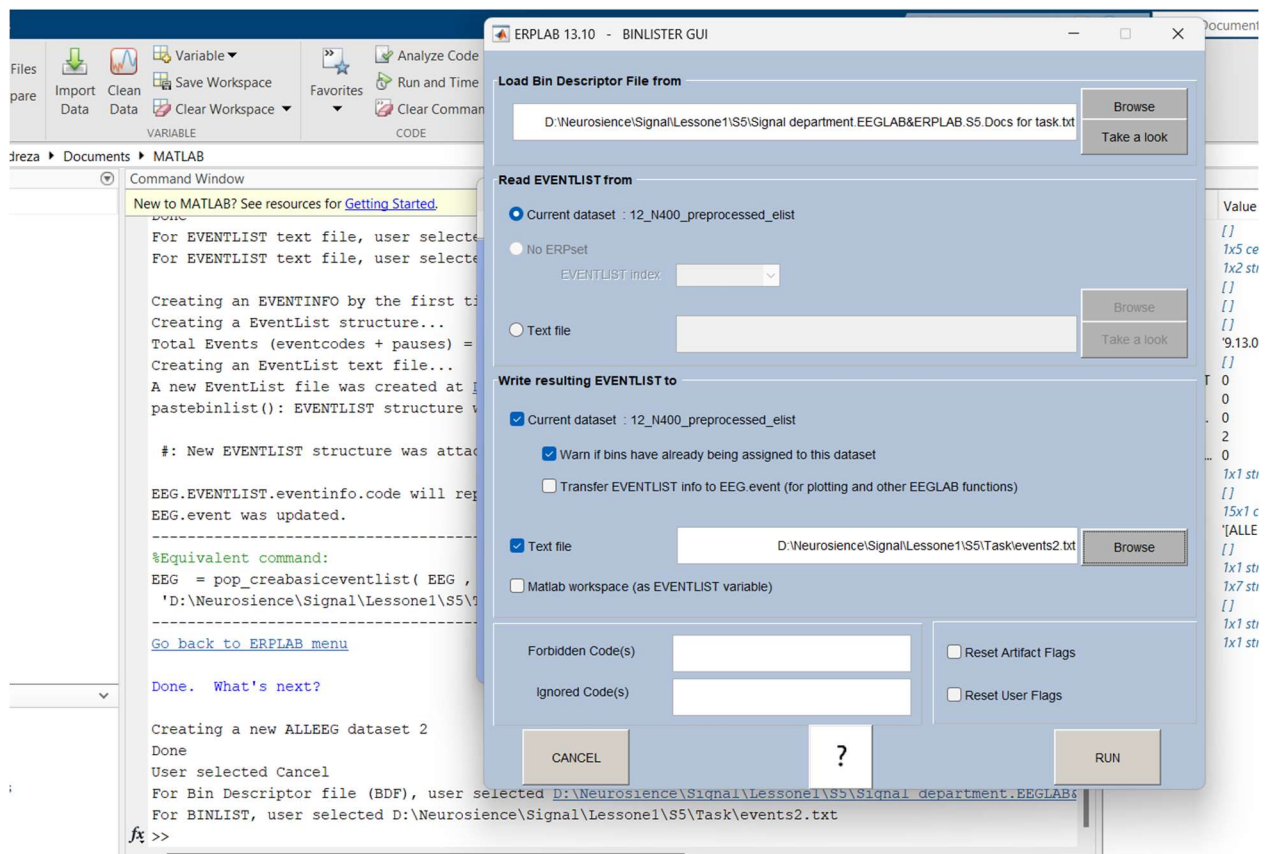
```
# Non-editable header begin -----
#
# data format.....: continuous
# setname.....: 12_N400_preprocessed_elist
# filename.....: Signal department.EEGLAB&ERPLAB.S5.Task Data.set
# filepath.....: D:\Neuroscience\Signal\Lesson1\55\
# nchan.....: 30
# srate.....: 76800
# npts.....: 200
# nevents.....: 363
# generated by (bdf).....:
# generated by (set).....: 12_N400_preprocessed_elist
# reported in.....:
# prog Version.....: 13.10
# creation date.....: 23-Sep-2026 20:30:11
# user Account.....:
#
# Non-editable header end -----
```

#	item	epoch	ecode	label	onset (sec)	diff (msec)	dura (msec)	b_flags (binary)	a_flags (binary)	enable	bin
1	0	202	--		6.1230	0.00	0.0	00000000	00000000	1	[]
2	0	202	--		7.7227	1599.61	0.0	00000000	00000000	1	[]
3	0	111	--		11.0555	4132.21	0.0	00000000	00000000	1	[]
4	0	211	--		13.0879	1232.42	0.0	00000000	00000000	1	[]
5	0	201	--		13.8350	747.07	0.0	00000000	00000000	1	[]
6	0	111	--		14.7881	953.12	0.0	00000000	00000000	1	[]
7	0	211	--		15.9375	1149.41	0.0	00000000	00000000	1	[]
8	0	202	--		16.3623	424.80	0.0	00000000	00000000	1	[]
9	0	121	--		17.5537	1191.41	0.0	00000000	00000000	1	[]
10	0	221	--		18.7197	1166.02	0.0	00000000	00000000	1	[]
11	0	201	--		19.1465	426.76	0.0	00000000	00000000	1	[]
12	0	121	--		20.5090	1355.47	0.0	00000000	00000000	1	[]
13	0	221	--		21.7852	1283.20	0.0	00000000	00000000	1	[]
14	0	201	--		22.8428	1057.62	0.0	00000000	00000000	1	[]
15	0	111	--		23.5674	724.61	0.0	00000000	00000000	1	[]
16	0	211	--		24.7178	1150.29	0.0	00000000	00000000	1	[]
17	0	201	--		25.0420	324.22	0.0	00000000	00000000	1	[]
18	0	111	--		26.3672	1325.20	0.0	00000000	00000000	1	[]
19	0	211	--		27.5996	1232.42	0.0	00000000	00000000	1	[]
20	0	201	--		28.3779	778.22	0.0	00000000	00000000	1	[]
21	0	111	--		29.3154	937.50	0.0	00000000	00000000	1	[]
22	0	211	--		30.4492	1133.79	0.0	00000000	00000000	1	[]
23	0	201	--		31.1943	745.12	0.0	00000000	00000000	1	[]
24	0	111	--		32.1318	937.50	0.0	00000000	00000000	1	[]
25	0	111	--		32.1146	1149.41	0.0	00000000	00000000	1	[]

- Find the response time for **row 5 (event code 201)**.
- Hint:* Response time = time difference between the target word and the response event.

$$13.0879 \text{ s} - 13.8350 \text{ s} = 747.07 \text{ ms}$$

2. Assign events to Bins using the *BDF_N400.txt* file.



- Report the number of events in each Bin.

SUMMARY:

Your events are NUMERICS

Below is the number of each event types:

```

Type 111 appears 30 times
Type 112 appears 30 times
Type 121 appears 30 times
Type 122 appears 30 times
Type 201 appears 114 times
Type 202 appears 9 times
Type 211 appears 30 times
Type 212 appears 30 times
Type 221 appears 30 times
Type 222 appears 30 times

```

****Below is the number and description of each bin:**

```

bin 1, # 60, Prime word, related to subsequent target word
bin 2, # 60, Prime word, unrelated to subsequent target word
bin 3, # 58, Target word, related to previous prime, followed by correct response
bin 4, # 55, Target word, unrelated to previous prime, followed by correct response

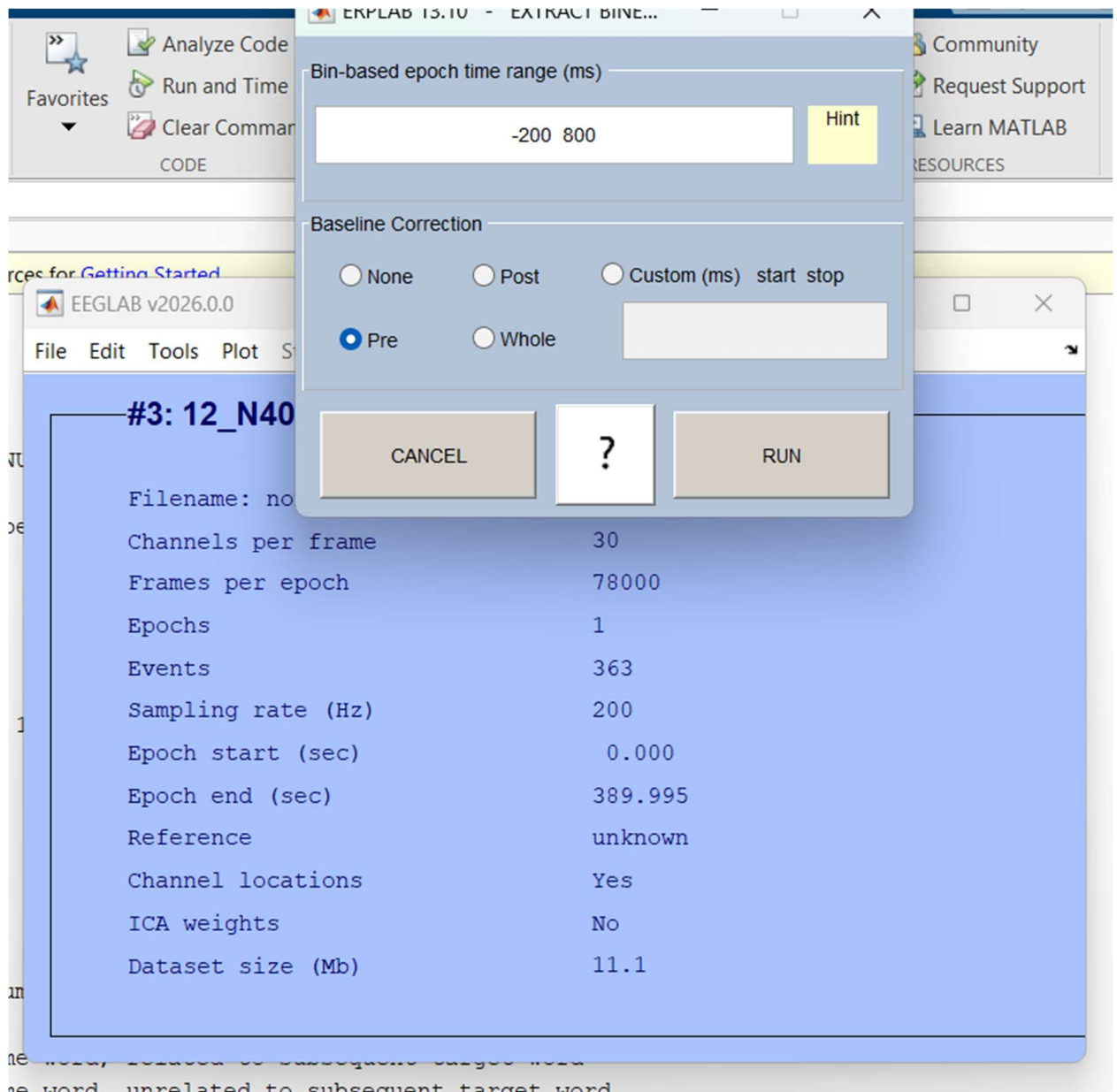
```

Done. What's next?

#3: 12_N400_preprocessed_elist_b...

Filename:	none
Channels per frame	30
Frames per epoch	78000
Epochs	1
Events	363
Sampling rate (Hz)	200
Epoch start (sec)	0.000
Epoch end (sec)	389.995
Reference	unknown
Channel locations	Yes
ICA weights	No
Dataset size (Mb)	11.1

3. Epoch the data and apply baseline correction



- Epoch range: -200 ms to 800 ms
- Baseline: pre-stimulus interval (-200 to 0 ms)

Sampling rate (Hz)	200
Epoch start (sec)	-0.200
Epoch end (sec)	0.795
Reference	unknown



4. **Compute averaged ERPs (grand average across epochs)** and report the results.

ERPLAB 13.10 - EEGset -> ERPset Averager

EEG Dataset(s) Index

4

Epochs to Include in ERP Average

☐ Include ALL epochs (ignore artifact detections)

☒ Exclude epochs marked during artifact detection (highly recommended)

☐ Include ONLY epochs marked with artifact rejection (be cautious!)

☐ Include ONLY the following epochs :
(epoch indices or filename (.txt))

Epoch Subset Assistant

☐ Use filename

☐ Use epoch indices

View file

Load list

clear editor

Save List as

☒ Exclude epochs that contain either "boundary" or invalid events (highly recommend...

Data Quality Quantification

☒ On - default parameters

☐ On - custom parameters

☐ No Data Quality measures

Set DQ options... ?

Power Spectra

Create a file with:

☐ Total Power Spectrum ?

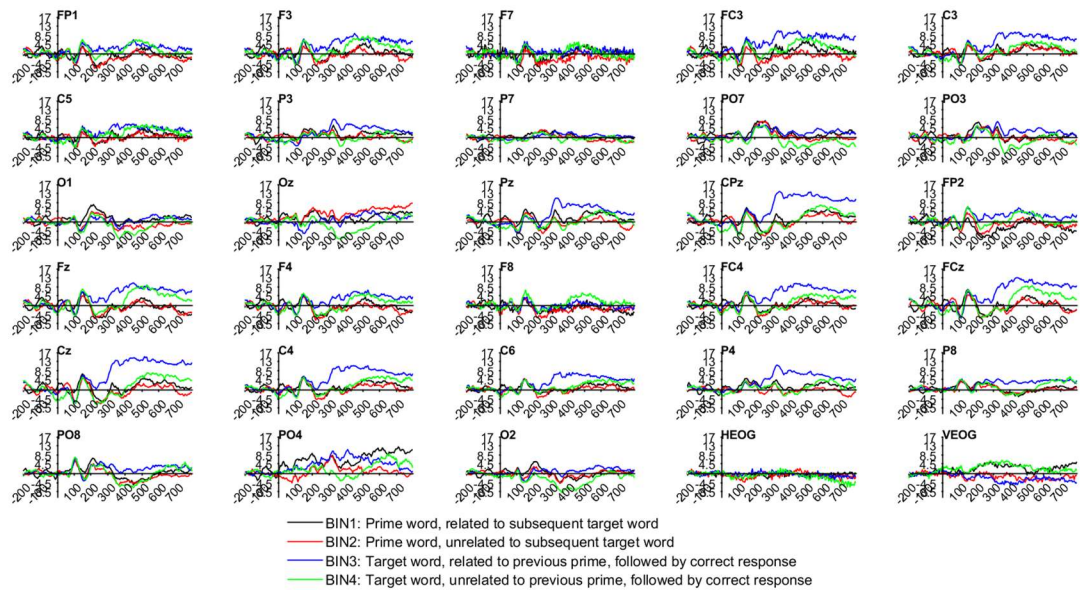
☐ Evoked Power Spectrum ?

☐ Tapering function - e.g. 'hanning' [-200 ...

CANCEL

?

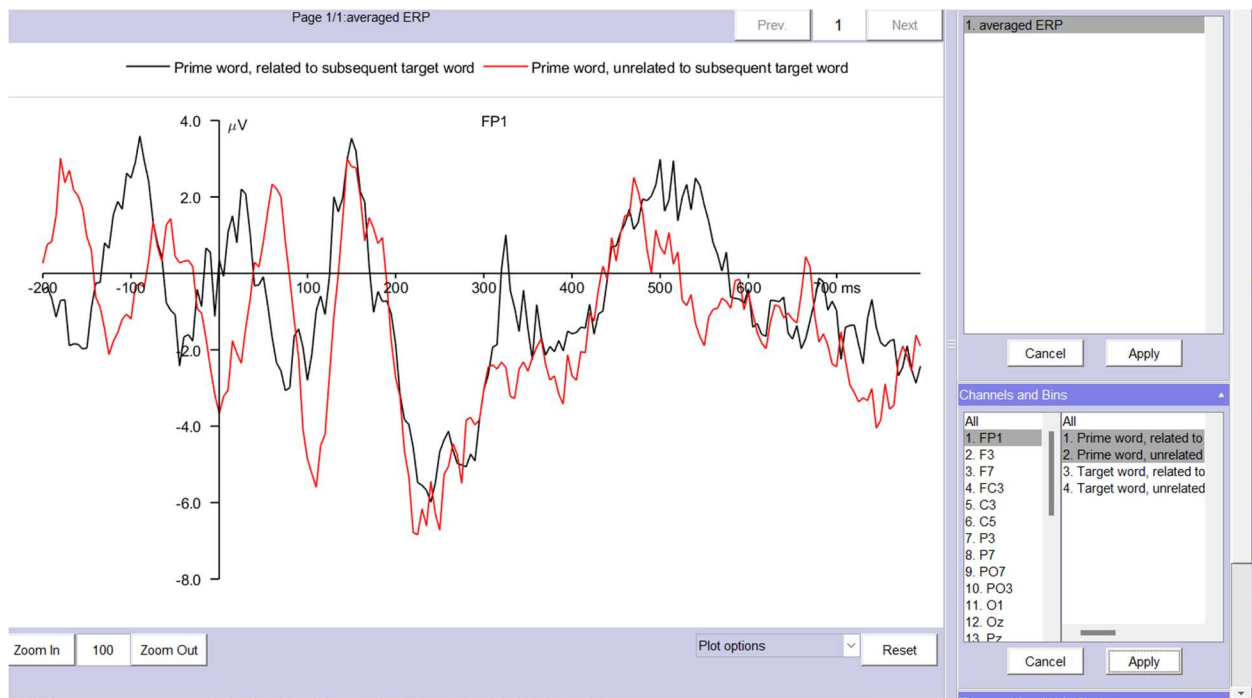
RUN



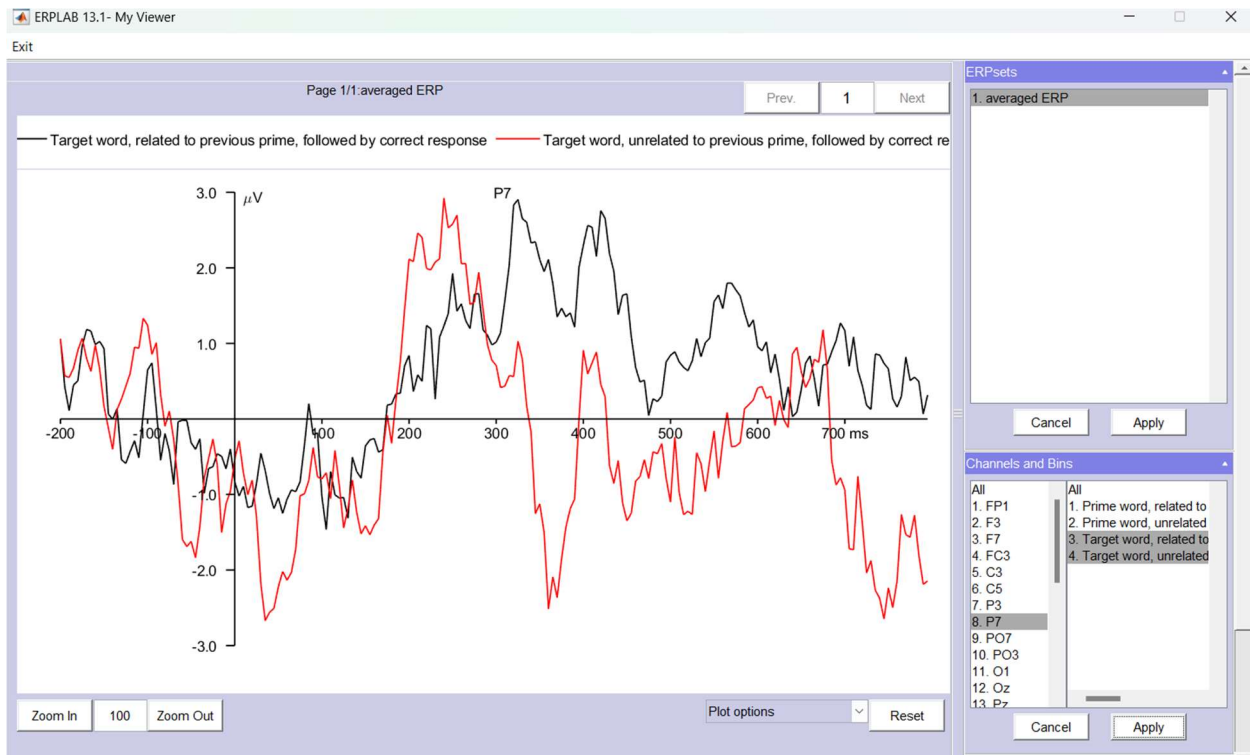
Averaged ERPs were computed for all bins. The resulting ERP waveforms were obtained for Bin 1, Bin 2, Bin 3, and Bin 4.

5. Plot ERPs:

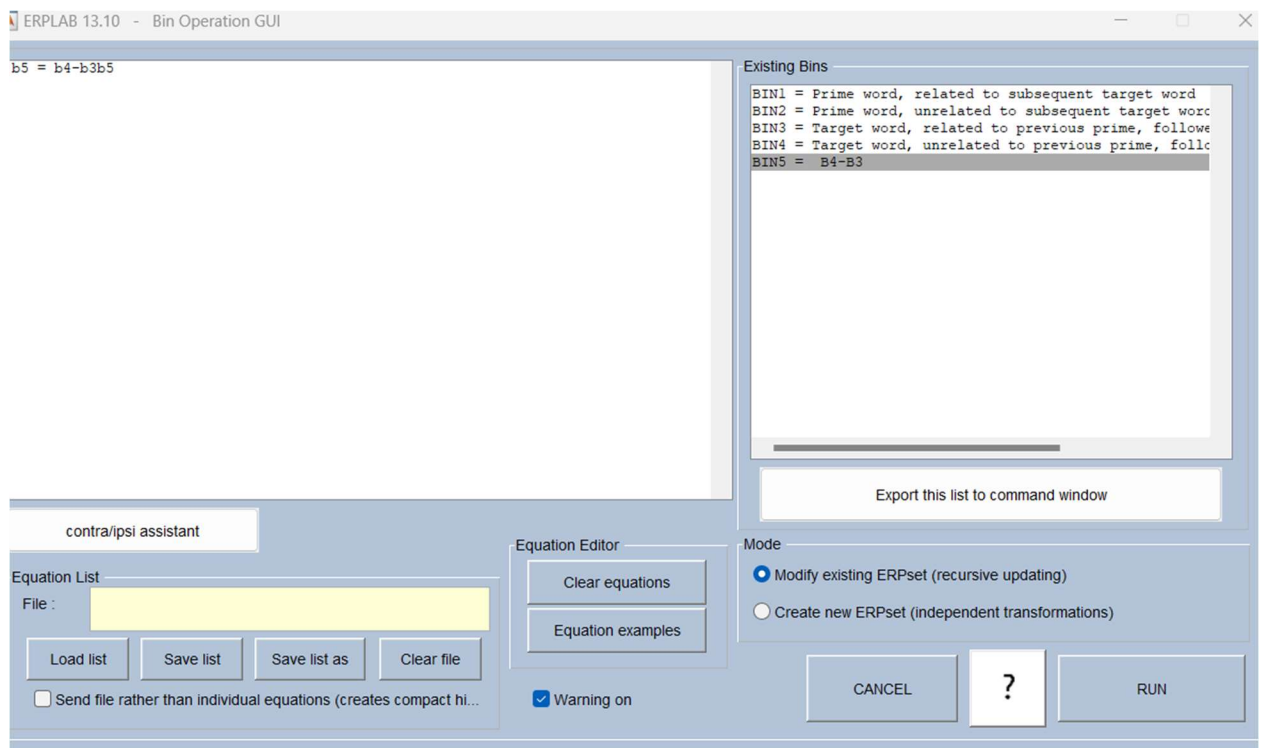
- Once for Bins 1 & 2 together



- Once for Bins 3 & 4 together



6. Create an additional Bin (Bin 5) defined as *Bin 4 – Bin 3*.

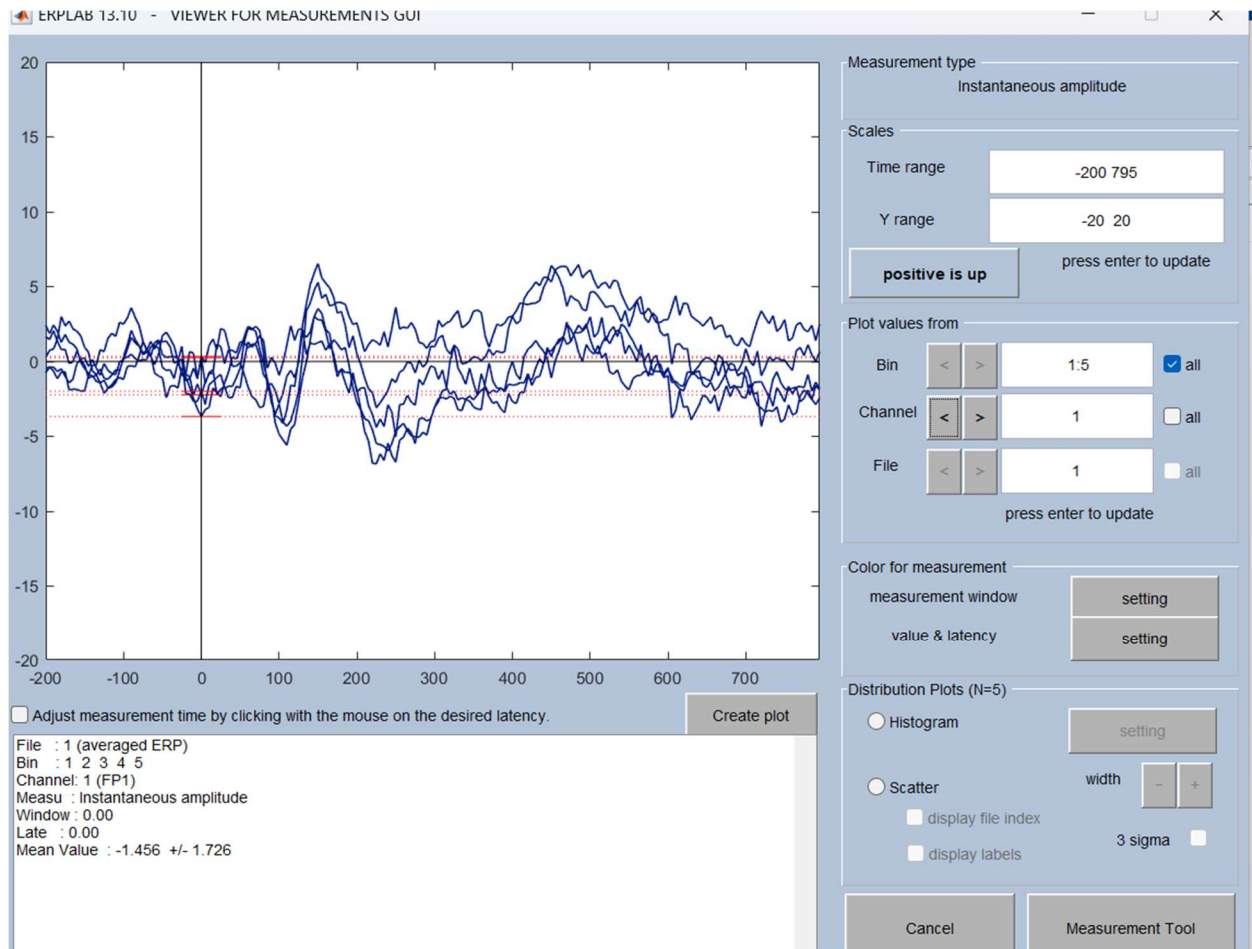


- Plot Bin 5.

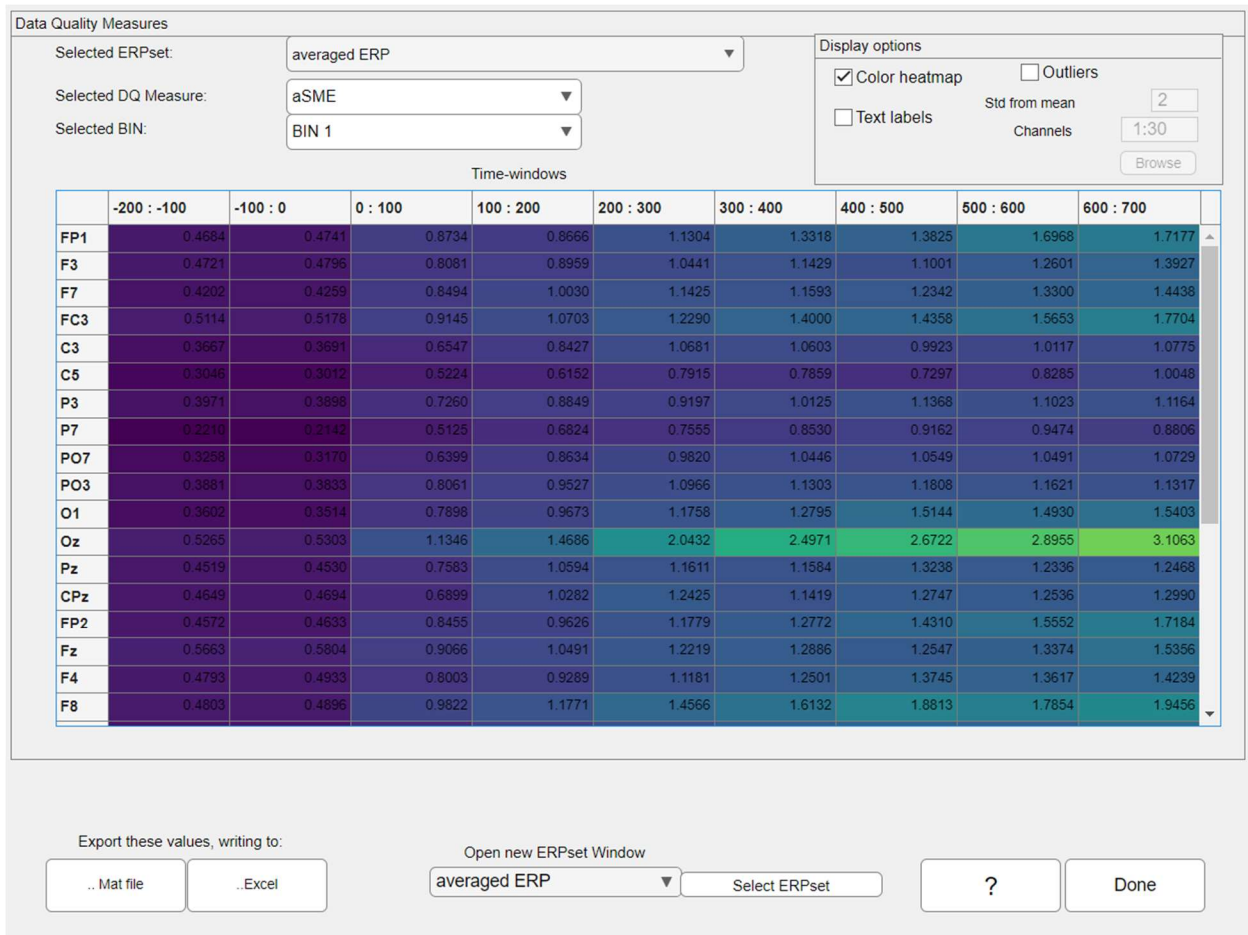


- Explain what information we can infer from this difference wave.

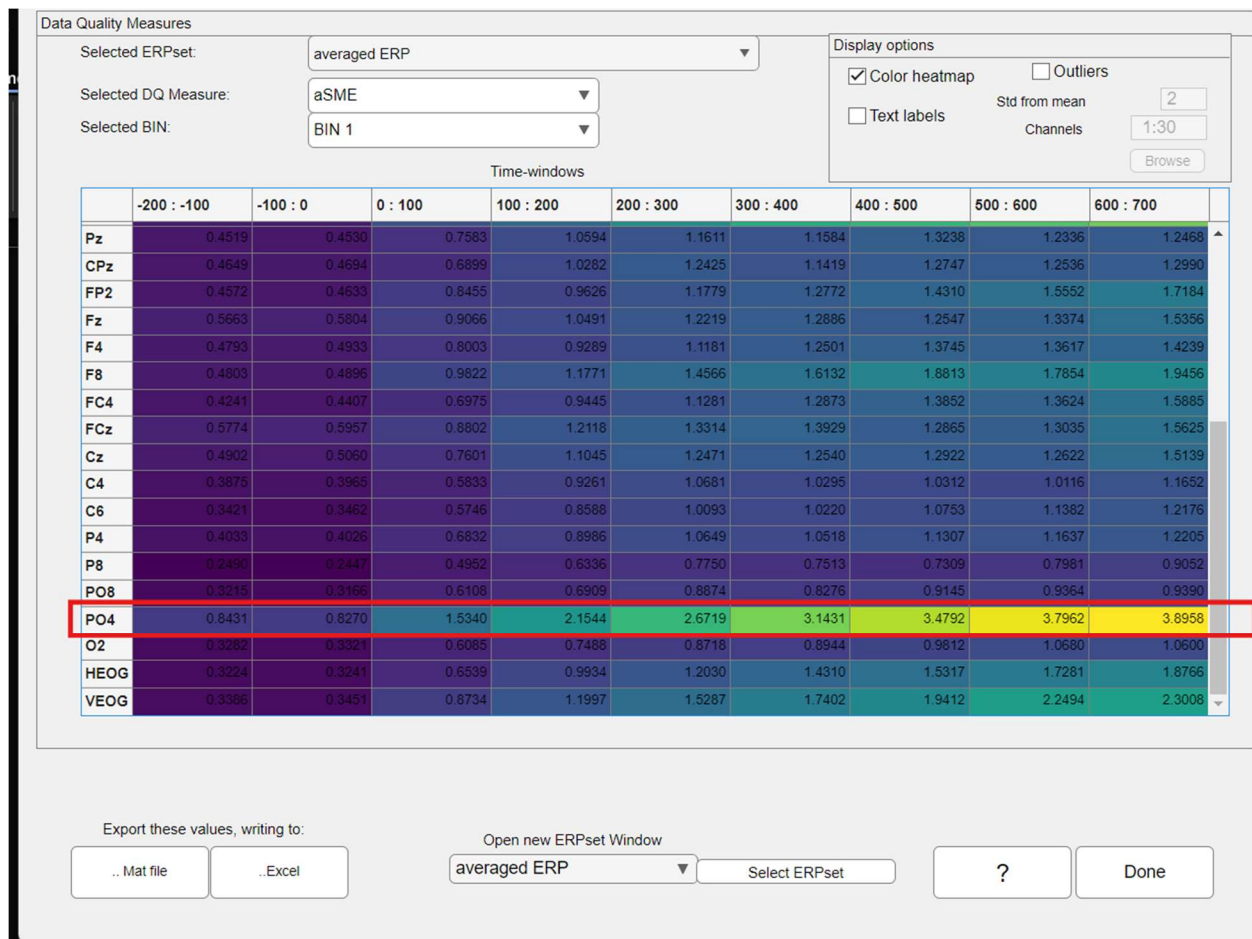
The difference wave shows the difference in brain responses between Bin 4 and Bin 3. It helps us identify condition-specific ERP components by reducing activity that is common to both conditions. It can also show differences in the amplitude and timing of ERP peaks.



7. Report the Data Quality Table (aSME) for your averaged epochs.



- Identify channels and time windows where the signal quality is poor.



channel PO4 showed relatively poor signal quality, particularly during the 300–700 ms time window, where the signal quality decreased noticeably.